

NUCLEAR PROCESSES

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<https://DenisPhysicsLab.github.io>



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NUCLEAR PROCESSES

SENIOR 4: TERM 3

Theme: Modern physics

TOPIC 6: NUCLEAR PROCESSES

20 PERIODS

Competency: The learner should understand how nuclear processes occur, their uses, and the dangers associated with them

LEARNING OUTCOMES The learner should be able to:	SUGGESTED LEARNING ACTIVITIES	SAMPLE ASSESSMENT STRATEGY
a. understand the processes of nuclear fission and fusion and the associated energy changes (u) b. understand the spontaneous and random nature of nuclear decay and interpret decay data in terms of half-life (k, u, s) c. know the applications of radioactivity and the dangers associated with exposure to radioactive materials. (k, u) d. understand and appreciate that there are significant social, political, and environmental dimensions associated with use of nuclear power. (u, v/a)	- In groups, learners research and use knowledge of atomic structure to explain in a presentation/on a poster: - the processes of nuclear fission and nuclear fusion and balance nuclear reactions - how energy is produced in a controlled way in a reactor - how nuclear energy is used - In groups, learners research and prepare a presentation on: - the penetrating and ionizing powers of different types of radiation - the decay of radioactive isotopes using balanced nuclear reactions - the meaning of half-life and how it may be used - background radiation and its origin - why waste containing radioactive isotopes with long half-lives presents a serious environmental problem - the medical and industrial uses of nuclear material - In pairs, learners research on the advantages and disadvantages of the use of nuclear materials with reference to world events; the regulations about the use and control of radioactive materials.	- Listen to group and pair discussions and observe activities, asking probing questions to promote critical thinking and deepen learning. - Assess the quality of posters and presentations to evaluate and accelerate progress towards the learning outcomes.
Note: a) Calculation of half-life using decay law equation is not required at this level. However, graphical method should be emphasized b) Details of how radiations damage the body are not required. However, occurrence of recent world nuclear accidents needs to be mentioned.		
ICT Support Learner watches computer simulations about thermionic emission and photoelectric effect.		

RADIOACTIVITY

This is the spontaneous and continuous disintegration of a heavy and unstable nucleus to form a stable nucleus with emission of radiations.

The radiations emitted are:

- 1) Alpha particles (α)
- 2) Beta particles (β)
- 3) Gamma rays (γ)

Radioactive nuclide (Radioactive substance)

This is a nuclide that undergoes spontaneous decay giving off alpha particles, beta particles and gamma rays.

Radioisotopes

These are isotopes that undergo spontaneous decay giving off alpha particles, beta particles and gamma rays.

Properties of each particle

Alpha particles

An alpha particle is a helium nucleus which is positively charged. ie ${}^4_2\text{He}$

It has a mass number of 4 and an atomic number of 2

It has a charge of +2

Properties

- It is positively charged.
- It is deflected by both electric and magnetic fields.
- They have the least penetrating power because of their relatively large mass.
- They have the greatest ionising power due to their heavy charge.
- They have a very short range in air and can be stopped by a thick piece of paper.

- They cause fluorescence.
- They affect a photographic plate.

Beta particle

A beta particle is a fast moving electron.

It has a charge of -1 and no mass number ie ${}_{-1}^0\text{e}$

Properties

- They are negatively charged.
- They are deflected by both electric and magnetic fields.
- They have higher penetrating power than alpha particles.

- They have less ionising power than alpha particles.
- They have a higher range in air than alpha particles and are stopped by a thin sheet of aluminium foil.
- They cause fluorescence.
- They affect a photographic plate.

Gamma rays

These are electromagnetic radiations of a very short wavelength and high frequency.

They have no mass number and no charge ie ${}^0_0\gamma$

Properties

- They have no charge.
- They are not deflected by both electric and magnetic fields.

- They have the greatest penetrating power.
- They have the least ionising power.
- They have the greatest range in air and can be stopped by a thick lead metal.
- They travel at a speed of light ($3.0 \times 10^8 \text{ ms}^{-1}$)
- They cause fluorescence.
- They affect a photographic plate.

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Comparison of alpha particles, beta particles and gamma rays

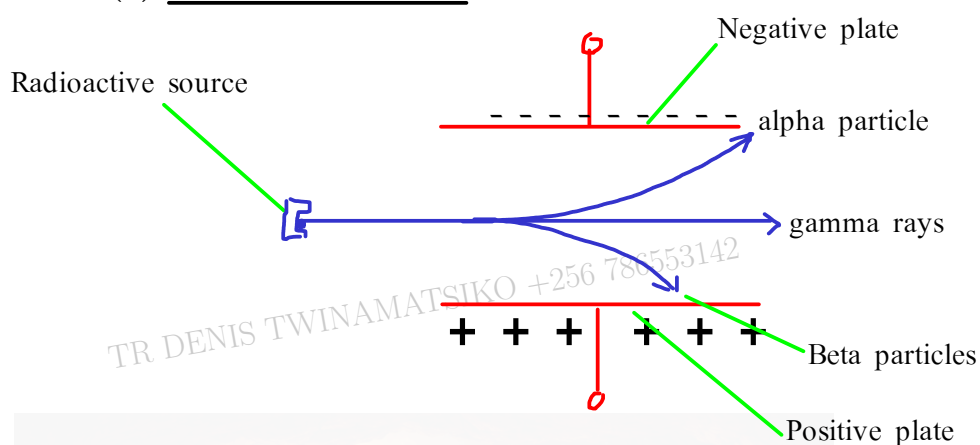


	Alpha particle	Beta particle	Gamma rays
Charge	positively charged	negatively charged	carry no charge
Deflection	Deflected towards negative terminal in an electric field and towards south pole in a magnetic field	Deflected towards positive terminal in an electric field and towards north pole in a magnetic field	Not deflected
Penetrating power	least penetrating power	higher than alpha particle	highest penetrating power
Ionising power	greatest ionizing power	less ionizing power	least ionizing power
Range in air	very short range in air	higher range than alpha particles in air	highest range in air
Definition	helium nucleus	fast moving electron	electromagnetic waves
Penetrating power	can be stopped by a thick piece of paper	can be stopped by a thin sheet of aluminium foil	can be stopped by thick lead metal

Similarities

They all cause fluorescence.

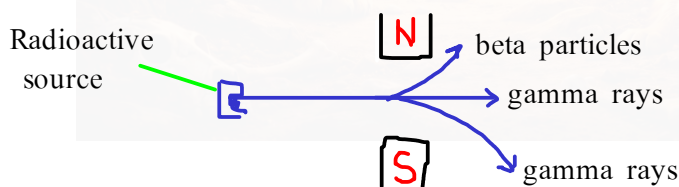
They all affect the photographic plate / film.

Deflection of alpha particles, beta particles and gamma rays**(a) In the electric field.**

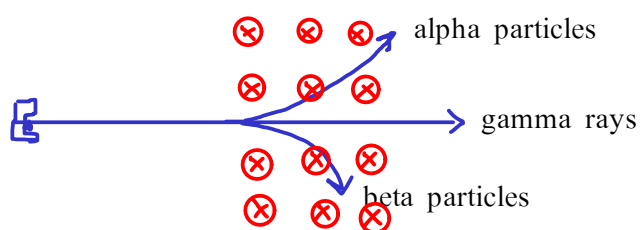
If radiations from a radioactive source are passed through a strong electric field, then:
Alpha particles are deflected towards the negative plate showing they are positively charged.

Beta particles are deflected towards the positive plate showing they are negatively charged.

Gamma rays are not deflected implying that they have no charge.

(b) In the magnetic field.

OR



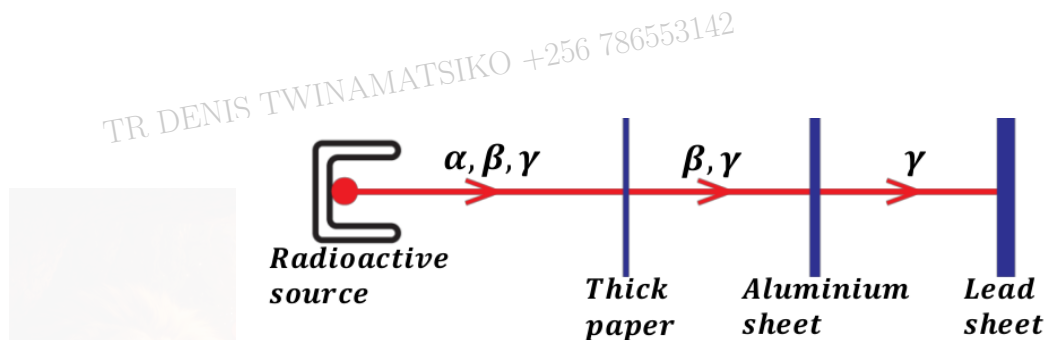
If radiations from a radioactive source are passed through a strong magnetic field:
Alpha particles are deflected towards the south pole which shows they are positively charged.

Beta particles are deflected towards the North pole indicating that they are negatively charged.

Gamma rays are not deflected implying that they carry no charge.

NB: Alpha particles are less deflected than beta particles implying that alpha particles are heavier than beta particles.

PENETRATING POWER OF RADIATIONS



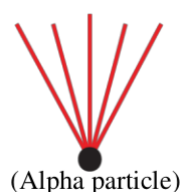
Alpha particles have the least penetrating power and can be stopped by a thick sheet of paper.

Beta particles have a greater penetrating power than alpha particles and can be stopped by a thin sheet of aluminium.

Gamma rays have the greatest penetrating power and can be stopped by a thick block of lead.

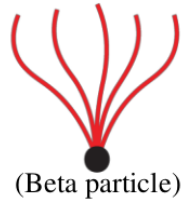
IONIZING POWER OF RADIATIONS (Nature of Radiations in air)

(i) Alpha particles



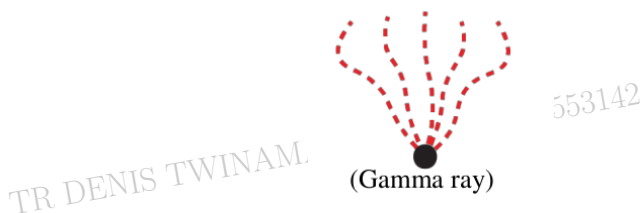
Alpha particles produce straight traces because they are heavy and they cause greater ionization of the air through which they pass.

(ii) Beta particles



Beta particles produce irregular and light traces.

(iii) Gamma rays



Gamma rays do not traces but leave hairy traces after colliding with the air molecules since they have a least ionization power.

RADIOACTIVE DECAY

Radioactive decay is the spontaneous breakdown of a radioactive nucleus with emission of alpha particles, beta particles and gamma rays.

Types of radioactive decay are:

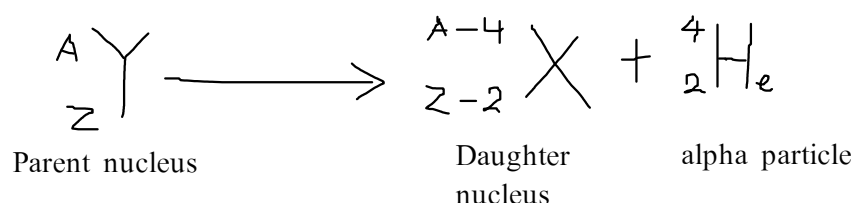
- 1) Alpha decay
- 2) Beta decay
- 3) Gamma decay

(1) Alpha decay

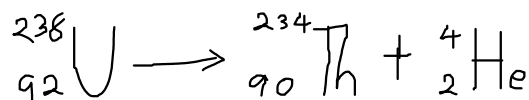
During alpha decay, a helium (alpha) particle is emitted.

A daughter nucleus with atomic number and mass number less than that of the parent nucleus by 2 and 4 respectively is produced.

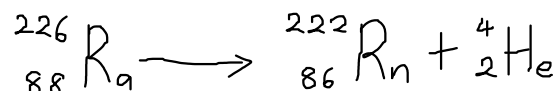
The equation as shown below.



eg Uranium-238 decays by emitting an alpha particle to become Thorium.

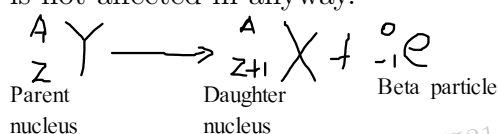


Radium decays by emitting an alpha particle to become Radon.

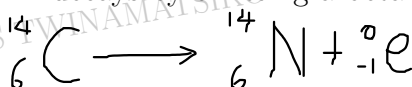


(2) Beta particle

During Beta decay, an electron (beta particle) is emitted. A daughter nucleus whose atomic number is greater than that of the parent nucleus by 1 is formed. The mass number is not affected in anyway.

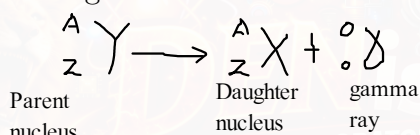


eg Carbon-14 decays by emitting a beta particle to become Nitrogen.



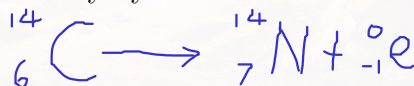
(3) Gamma decay

During gamma decay, a new element with atomic and mass number similar to those of the original element is formed.

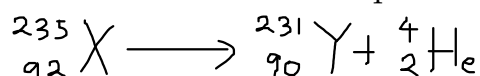


Examples

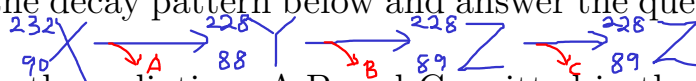
- Carbon-14 decay by emission. Write an equation to show the decay.



- A radioactive nuclide, ${}^{235}_{92}\text{X}$ emits an alpha particle and a new nuclide Y is formed. Write a balanced equation.



- Study the decay pattern below and answer the questions that follow.



- Identify the radiations A, B and C emitted in the decay process.
- State two differences between radiations A and B

HALF LIFE ($t_{\frac{1}{2}}$)

Half life is the time taken for half a number of atoms in any gives sample of a substance to decay.

OR It is the time taken for half the initial amount (mass) of a substance to decay.

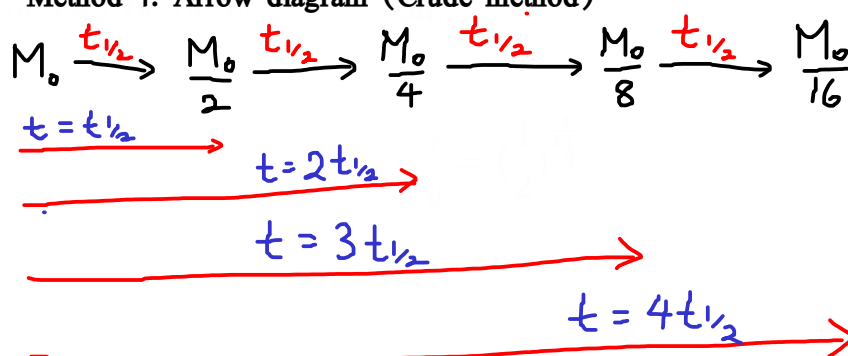
Activity (Count rate)

This is the number of radioactive particles emmitted per second.

NB: The half life of any radioactive substance can be measured in any unit of time ie milli seconds, micro seconds, seconds, minutes, hours, weeks, months and years.

Determining half life

Method 1: Arrow diagram (Crude method)



where

M_t is Mass remaining

M_0 is initial mass

t is time taken

$t_{\frac{1}{2}}$ is half life

Method 2: Graphical method

(1.2)

Where $\frac{t}{t_{\frac{1}{2}}} = n$ is number of half lives

Examples

1. The half life of a certain radioactive isotope is 24 days. If the initial mass 0.64 g. Find

(a) The mass remaining after 120 days

(b) The mass decayed after 120 days

(c) How long it will take to have only 0.04 g remaining undecayed

(d) The fraction of the radioisotope:

(i) that remains after 120 days

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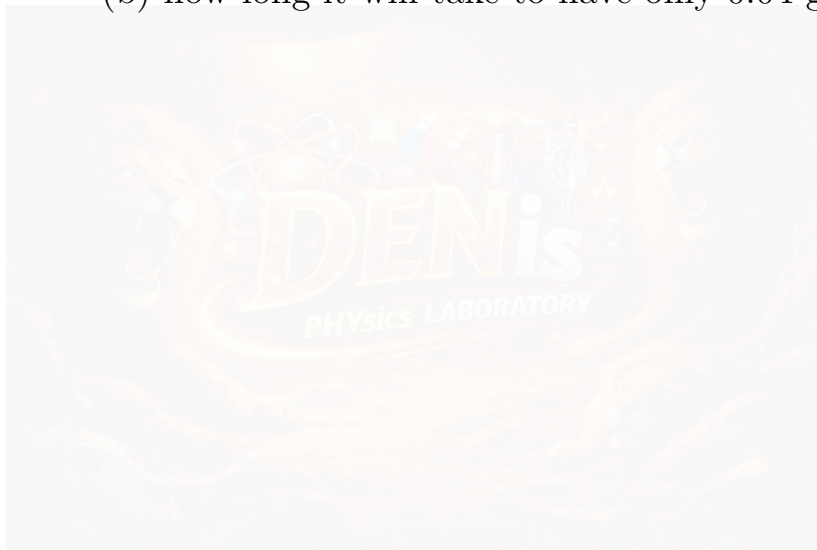
(ii) that will have decayed after 96 days.

2. The half life of Uranium is 24 days. Calculate the initial mass of the sample of Uranium if 3 g of it remains after 120 days.

3. The half life of certain radioisotopes is 24 days. If the initial mass is 0.64 g. Find:

(a) the mass remaining after 120 days.

(b) how long it will take to have only 0.04 g remaining undecayed.



Trial questions

1. A radioactive element has a half life of 9 minutes. Given that the original count rate is 256 counts per minute, find:

(a) The time taken to reach a count rate of 16 counts per minute?

(b) What fraction of the original number of atoms of the radioactive element will be left by the time the count rate is at 16 counts per

minute?

2. The half life of Cobalt - 60 is 5 years. What fraction of the isotope remained after 15 years?

3. A mass of a radioactive substance decays to $\frac{1}{16}$ of its original value after 36 days.

(a) Find its half life

(b) What fraction of the original mass will have decayed after 36 days.?

4. A radioactive element of mass 9.6 g has a mass of 0.15 g after 24 hours.

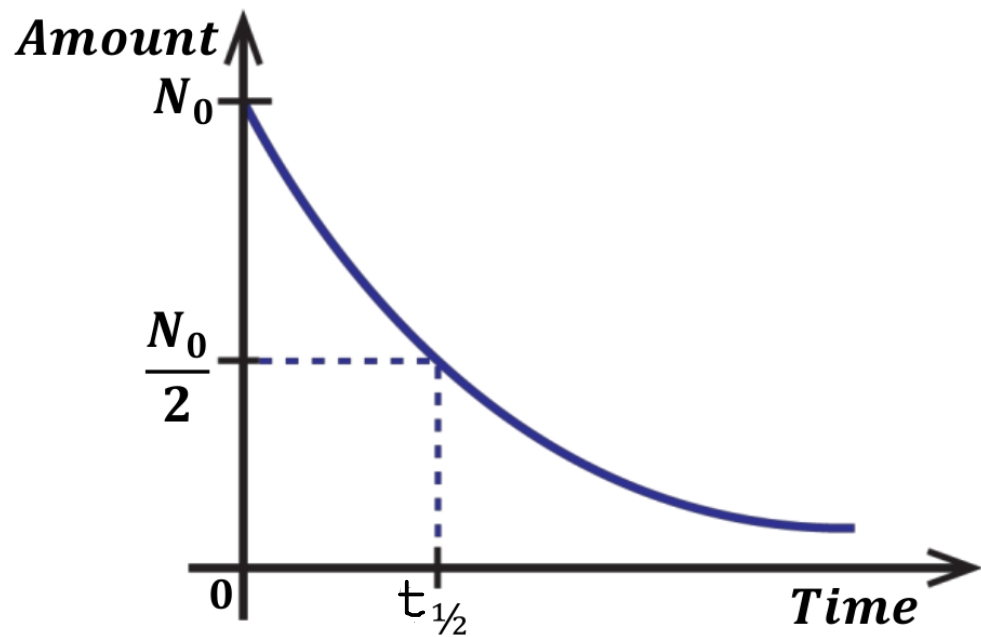
(a) Find the half life of this element.

(b) Mass decayed

GRAPHICAL METHOD OF DETERMINING HALF LIFE

The following steps are followed:

- A graph of activity/ count rate, N of an element against time, t is plotted.
- A horizontal line from half of the original amount to meet the curve is drawn.
- A vertical line from the point on the curve to meet the time axis is drawn.
- The half life is read from where the vertical line meets the time axis.



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EXAMPLES

1. The table below shows the count rate of a radioactive sample at different time intervals.

time (minutes)	0	40	80	120	160
count rate(/min)	1000	570	330	190	130

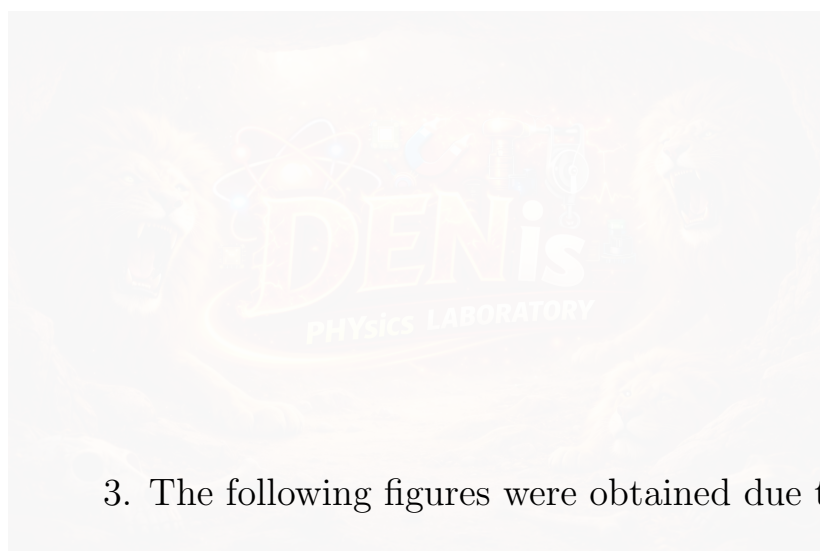
- (a) Plot a graph of count rate against time

(b) Use the graph to obtain the half life of the substance

2. Study the table below and use it to find the half life of the substance.

time (seconds)	0	5	10	15	20
count rate (/second)	250	175	76	38	25

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3. The following figures were obtained due to ignition of radon gas.

time (minutes)	0	102	155		300
count rate (min^{-1})	1600		200	100	50

(a) (i) Plot a graph of count rate against time.

(b) (ii) Determine the half life.

(iii) Find the missing values.

- (c) (i) What is the count rate after 200 minutes?
(ii) After how many minutes is the count rate 100 per minute?

APPLICATIONS OF RADIOACTIVITY

(a) Industrial uses:

They are used;

In tracer techniques to investigate the flow of liquids in chemical plants.
(Identifying oil leakages in oil pipe lines).

In the automatic control of thickness or uniformity of materials in industries. (e.g Cigarettes)

In the study of wear and tear in machinery.

To detect faults in thickness of metals sheets in welded joints. (gamma rays)

In food preservations:

to sterilize equipments in food industry

to detect smoke

in energy production: Nuclear reactors use radio isotopes to produce electricity.

(b) Medical uses

Used to treat cancerous cells. (Radiotherapy).

They are used to kill bacteria in food (x- rays).

Used to sterilize medical surgical equipments.

Locate broken bones

Asses the amount of blood in a patient

(c) Archeological uses

Used to determine the time that has elapsed since death of a certain

organism occurred in a process called carbon dating.

(d) **Geology**

They are used to determine the age of rocks.

(e) **Biological uses**

used to study the uptake of fertilizers by plants.

Used to sterilize insects and hence eliminate pests that destroy crops.

(f) **Defence**

Nuclear reactions of fusion and fission are used in manufacture of weapons of mass destruction like nuclear and atomic bombs.

**DANGERS ASSOCIATED WITH EXPOSURE TO
RADIOACTIVE MATERIALS**

- They cause genetic mutation
- They cause radiation burns
- They destroy living cells
- They damage eye sight
- They cause cancer.
- They cause sterility

**SAFETY PRECAUTIONS WHEN HANDLING
RADIOACTIVE SUBSTANCES**

- They should be held with forceps or a pair of tongs and not with bare hands.
- Avoid eating, drinking or smoking where radioactive sources are in use.
- Radioactive sources must be kept in lead boxes when not in use.
- Wash hands thoroughly after exposing them to radioactive materials.
- Any cut on the body should be covered before dealing with radioactive sources.
- Operators should put on gloves and lead coats.
- During experiments with radioactive materials, the radiations should not be directed towards the people.

BACKGROUND RADIATIONS

These are ionizing radiations from a variety of natural and artificial sources that are always present in the environment.

Natural sources of background radiations

Natural sources of background radiation include the following

Cosmic Radiation

These are radiations that reach the Earth from space.

The sun is a major source of cosmic rays. Cosmic rays originating from the sun, stars and other major events in outer space are continuously striking the Earth.

The majority of these cosmic rays are absorbed by the Earth atmosphere but the more energetic radiations interact with the atoms in the atmosphere creating energetic neutrons.

Terrestrial Radiation

These are radiations from radioactive materials that exist naturally in soil and rock.

E.g. radiations from uranium, thorium, and radium. Some rocks are radioactive and give off radioactive radon gas. Essentially all air contains radon. In addition, water contains small amounts of dissolved uranium and thorium, and all organic matter (both plant and animal) contains radioactive carbon and potassium.

Some of these materials are ingested with food and water, while others (such as radon) are taken in.

Internal Radiation

These are radiations from radioactive materials which are present in the human body.

These come from natural radioactive sources such as Carbon-14 in the air we breathe. Fruits and crops take in radioactive materials from the soil as well as Carbon-14 from the air.

Artificial sources of background radiations

These are mainly from human activities.

They include:

Nuclear power stations

Major incidents from nuclear power stations have released radiations into the environment.

Nuclear waste from power station also accounts for a proportion of artificial background radiation.

Nuclear weapons

Nuclear weapon testing resulted in an increase of radiation in the environment because of radioactive fallout from nuclear weapons testing.

Medical sources

Humans are exposed to radiations by medical procedures such as x-rays and radiotherapy.

Nearly all artificial background radiation comes from medical procedures such as receiving X-rays for X-ray photographs.

Detecting the radiation

The following can be used to detect radiations;

Photographic film

Photographic film goes darker when it absorbs radiations.

The more radiation the film absorbs, the darker it becomes.

People, who work with radiation, wear film badges which are checked regularly to monitor the levels of radiation absorbed.

Geiger-Muller tube and Cloud Chamber

The Geiger-Muller tube or cloud chamber detects radiation and each time it absorbs radiation, it transmits an electrical pulse to a counting machine.

This makes a clicking sound or displays the count rate.

The greater the frequency of clicks, the higher the count rate and the more radiation the Geiger-Muller tube or cloud chamber is absorbing.

NUCLEAR REACTIONS

A nuclear reaction is the process in which energy is produced by either splitting a heavy nucleus or combining two light nuclei.

Types of nuclear reactions

1. Nuclear fission
2. Nuclear fusion

1. NUCLEAR FISSION

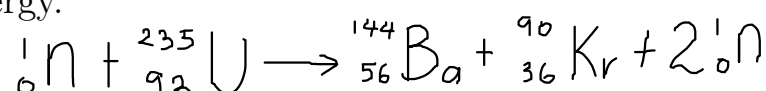
This is the splitting of a heavy nucleus into two lighter nuclei with release of energy.

Nuclear fission is initiated by bombarding an atom with a neutron.

In the process, a large amount of energy is released. This energy is called nuclear fission energy. Nuclear fission takes place in the **nuclear reactor**.

Example

Splitting of Uranium- 236 to form Barium and Krypton with release of energy.



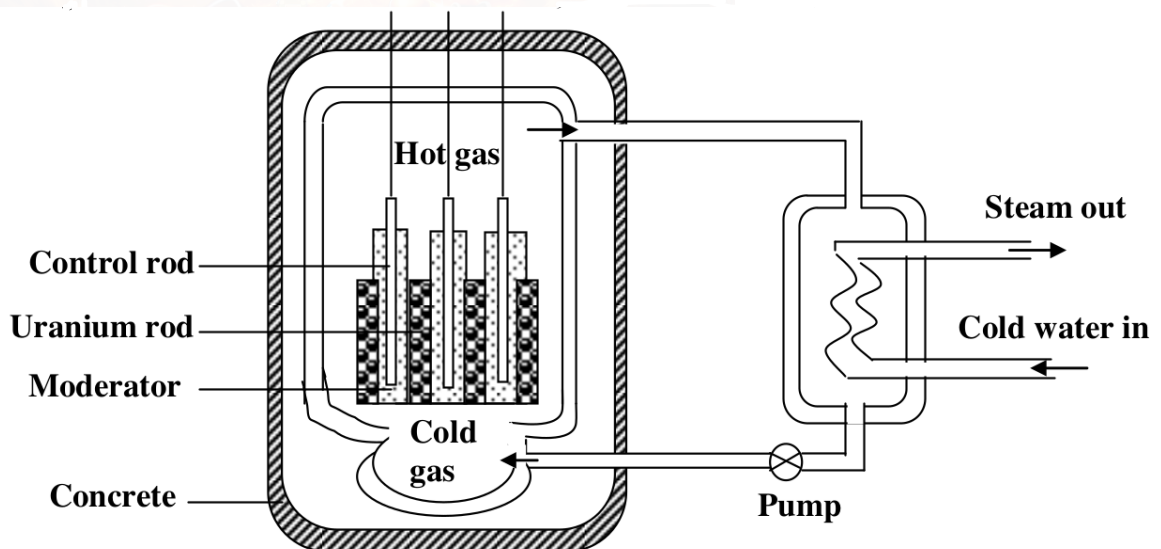
CONDITIONS FOR NUCLEAR FISSION TO OCCUR

1. Presence of heavy and unstable nucleus
2. Low temperature

Applications

1. Generation of heat
2. Generation of electricity
3. Manufacture of nuclear weapons eg bombs.

PRODUCTION / GENERATION OF ELECTRICITY IN A NUCLEAR REACTOR USING Uranium-235



The uranium rod undergoes fission and fast neutrons are produced as a result.

The neutrons collide with other atoms and the energy possessed by the neutrons changes to heat energy.

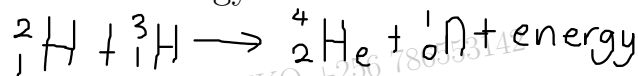
The heat energy is then absorbed by the coolant and is made to heat water to produce steam.

The steam then drives the turbines connected to a generator and electricity is produced.

NUCLEAR FUSION

This is the process where two lighter nuclei combine to form one heavy nucleus with release of energy.

Eg When two hydrogen isotopes combine to give helium and a neutron with release of energy.



The energy produced in this reaction is called nuclear fusion energy.

It takes place in the sun

CONDITIONS FOR NUCLEAR FUSION TO OCCUR

Presence of two lighter nuclei

Very high temperatures

USES/ APPLICATIONS OF NUCLEAR FUSION

1. Manufacturer of nuclear weapons
2. Generation of electricity
3. Generation of heat

Qn : (a) State the energy changes that occur in a nuclear reactor.

Nuclear energy $\rightarrow$ heat energy $\rightarrow$ kinetic energy $\rightarrow$ electrical energy

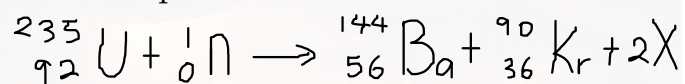
(b) State the difference between nuclear fission and nuclear fusion

Nuclear fission	Nuclear fusion
Is the splitting of a heavy nucleus into two lighter nuclei	Is the combining of two lighter nuclei to form a heavy nucleus
Requires a low temperature	Requires a high temperature
Requires neutrons for bombardment	Neutrons are not required
Results into four products	Results into three products
Energy released is high	Energy released is low.

SIMILARITIES

In both, nuclear energy is produced

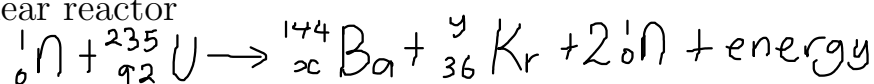
Qn In the equation



(a) State the type of reaction shown above

(b) Identify the particle X

Qn The equation below shows the reaction which takes place in the nuclear reactor



(a) Find the values of x and y

(b) State the use of nuclear energy

SOCIAL, POLITICAL AND ENVIRONMENTAL DIMENSIONS OF NUCLEAR POWER

(a) SOCIAL DIMENSIONS

Nuclear plants create jobs which improve people's way of living
Generation of electricity

(b) ENVIRONMENTAL DIMENSIONS

Radioactive wastes can remain radioactive and become dangerous to human health for a very long time

Nuclear power plants constantly emit radiations into the environment which can be dangerous when there is unnecessary exposure

Nuclear power generation can cause climatic change

(c) POLITICAL DIMENSIONS

Nuclear weapons and materials can influence international relations

Possession of nuclear materials can create tensions between nations fueling conflicts and rivalries

Nuclear materials can impact a nation's security influencing defense strategies and military planning

END

(b) State the use of nuclear energy

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END

SAMPLE ITEMS

UNEB 2025

The government has identified land in a certain area for establishing a nuclear power plant. As a standard procedure, the government has informed the people living in that area that they have to be relocated, since they cannot co-exist with the nuclear power plant. However, majority of the people have insisted that they cannot leave their land and think they can co-exist with the nuclear power plant. The

government has promised that after a specific period of time the community members will be allowed to return.

Due to this, data about the activity of the radioactive materials to be used in the nuclear power plant has been provided to the people in table 1.

Table 1

Count rate (per minute)	800	525	335	215	150
Time (1×10^9 years)	0	4	8	12	18

Unfortunately, none of the community members can make sense of the data.

The community members also want to know if there are any benefits of radioactive materials, in general.

You have been contacted by the community members for help.

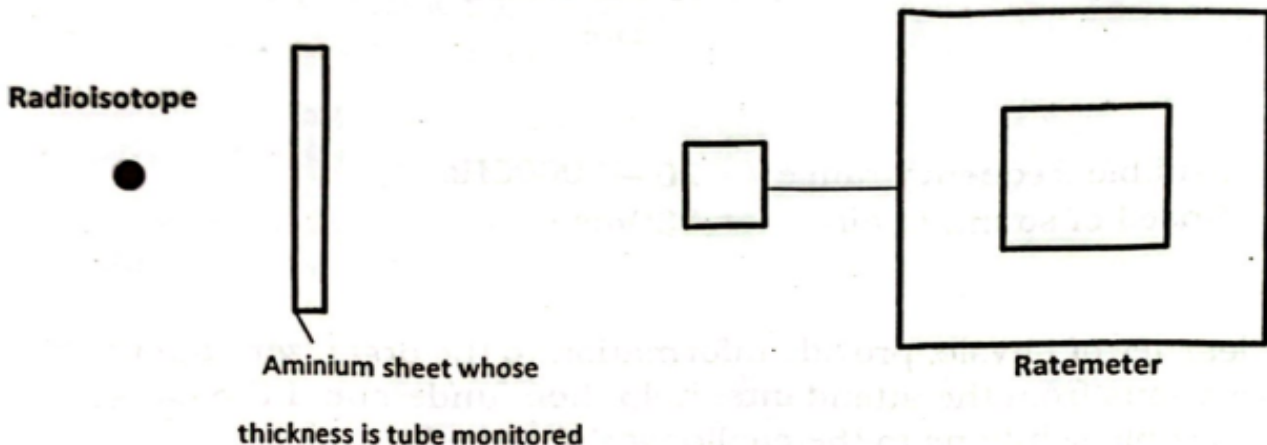
Hint:

The environment would be safe from waste radiation when the count rate falls below 35 counts per minute.

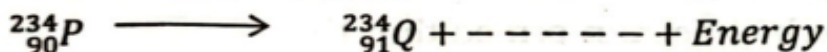
Task:

- (a) Help the community members to;
- understand the need for them to be relocated as soon as possible.
 - determine the time that has to pass before they can return to settle around the power plant again (use the data in table 1). $\times$
- (b) Describe any **two** benefits the community members are interested in.

A factory that produces Aluminium saucepans wishes to supply its customers according to the thickness of the materials they need. The operator of the machine for this purpose has thought of establishing a chart to be used to determine different thicknesses of Aluminium saucepans. He has been advised to use a radioisotope that emits suitable particles for this cause.



The operator doesn't know how the process of thickness monitoring is done. He has been provided with the radioisotope sample as show by the decay equation



The operator is not sure whether the radioisotope will be appropriate for use, and if there is any other application of this radioisotope. He has been cautioned to take precautionary measures to reduce on dangers associated with the use of radioisotope, only that he is not aware. To develop the chart, the operator has been told to take note of the background radiation of 60 counts per minute and use count rate for different thickness of Aluminium already established in order to easily produce for different thicknesses as required by different customers. He doesn't know what background radiation is and its sources.

Table below shows results obtained

Thickness of Aluminium(mm)	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0
Estimated count rate (Counts per minute)	560	310	180	120	90	75	60	60
Actual count rate (counts per minute)	---	---	---	---	---	---	---	---

A new customers has brought in new order to have Aluminium saucepan of thickness 3.6mm and the operator needs to set the machine to match the count rate, he has little knowledge about it.

Task

As a learner of physics, provide all the necessary information required by the operator to address all the challenges being faced.

END